

Machine Learning-Based Crop Yield Prediction Using Environmental and Agricultural Factors

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Abstract—Predicting agricultural crop yield is essential for ensuring food security and optimizing resource allocation in precision agriculture. This study proposes a robust computational framework for yield estimation by leveraging a multi-parametric dataset encompassing soil chemical properties (Nitrogen, Phosphorus, Potassium), climatic variables, and geographical factors. Four high-performance machine learning architectures—XGBoost, CatBoost, Random Forest, and Gradient Boosting—were implemented and evaluated on a dataset of 10,000 samples. The experimental results demonstrate that all models achieved a Coefficient of Determination (R^2) exceeding 0.9970. Specifically, the CatBoost Regressor emerged as the most efficient model, attaining a peak R^2 score of 0.9980 and a minimum Mean Absolute Error (MAE) of 0.8524. A stacking-based learnable fusion approach was also integrated to harmonize the predictive strengths of forest-based and boosting-based branches. The findings underscore the critical impact of Nitrogen levels and Rainfall patterns on yield variance, providing a scalable decision-support tool for precision farming and sustainable agricultural management.

Index Terms—Precision Agriculture, Crop Yield Prediction, Machine Learning, CatBoost, XGBoost, Random Forest, Sustainable Farming, Predictive Analytics.

I. INTRODUCTION

Agriculture is the foundation of the worldwide economy, but it has never been as threatened by climate change, unpredictable weather patterns, and the current growing pressure to feed the world. Tomato production is one of the crops that are very sensitive to environmental variables, ensuring that the farmers, stakeholders and policy makers require proper forecasting of yields. Historically, crop yield prediction has been based on statistical models which in most cases were unable to reflect the non-linear relationships that existed among the factors in agriculture. But with the introduction of Machine Learning (ML) and Deep Learning (DL), this field has been transformed to offer powerful tools to handle various types of data to make accurate estimates [1], [2].

The new developments in the field of computational intelligence shifted the emphasis of computational intelligence to the data-driven approaches. Scholars have managed to incorporate environmental factors (temperature, rainfall, and sunshine) and soil properties to improve predictability levels [10], [11]. Although classical models, such as Random Forest and Support Vector Machines were predominant over the last decade, new architectures, including Deep Neural Networks

(DNN) and Transformers are now establishing new standards in processing temporal and spatial agricultural data [5], [6]. Moreover, it can be integrated with the Internet of Things (IoT) sensors that enable real-time tracking of the health of crops, which, when combined with ML, offers a proactive method of managing yield.

Though the advancements have been made, yield prediction is a complicated issue because of the stochastic weather and the complex interplay of the genotypes and environmental stressors. The recent studies are also looking more into Explainable AI (XAI) as a means of not just making predictions but also comprehending the force behind productivity. In this paper, I present a machine learning-based model of predicting crop yields, in this case, environmental and agricultural variables. This paper will attempt to give a solution that is scalable to optimize tomato production by using modern algorithms and detailed datasets in the background of precision agriculture [13], [15].

- **Combination of Multi-source Agricultural Factors:** In contrast to most of the previous studies that consider either weather, or soil-based information, this study theorizes a comprehensive framework that combines soil chemical properties (N, P, K), physical environmental parameters (humidity, rainfall, sunlight), and agricultural inputs (fertilizers and pesticides) to enhance the power of prediction to a greater degree, as compared to many preced
- **Performance Optimization through Modern Architectures:** We test and compare various machine learning and deep learning architectures, namely how they can deal with non-linear interactions among different features to create a high-accuracy benchmark of tomato yield prediction in precision agriculture [3], [15].

II. LITERATURE REVIEW

Recent advances in computational agriculture have put machine learning as a basis of sustainable crop management. This chapter is a synthesis of the existing literature on predictive modeling, especially the development of algorithms and technology as well as the incorporation of various environmental and soil-based data.

- **Systematic Frameworks and Trends:** Van Klompenburg et al. [1] have presented a review of yield prediction in

which they found out that the most significant variables are the climatic variables such as temperature and precipitation. According to their study, Random Forest and SVM had been the most common classical models since they are trusted when handling non-linear data.

- **Advances in Deep Learning:** With the shift to high-dimensional data, deep architectures were adopted. Khaki and Wang [2] developed a Deep Neural Network (DNN) model that was more successful than the traditional models because it represented the complex interactions between genotypes and the environment, and neural networks have to be scaled to attain accuracy in agriculture.
- **Combination of Soil and IoT Data:** Sekhonne ny weather-e modern research. As pointed out by Chlingaryan et al. [13] the crops susceptible to the nutrient like tomatoes are sensitive to nitrogen (N) and soil pH. In addition, Hossain et al. [4] integrated real-time data of IoT sensors with gradient boosting algorithms, thereby reducing the error rates in the local yield prediction to a minimum.
- **Feature Engineering and Inputs:** The multi-source data, such as the one used by Anshumish, has demonstrated that the use of farming inputs, such as the application of fertilizer and pesticides, and the moisture content of the soil can give a better picture of the growing cycle, and therefore even better accuracy in prediction.
- **Interpretability and Future Directions:** We are moving towards Explainable AI (XAI). Chen et al. [8] proposed that farmer trust is based on transparent ML models which allow farmers to understand how specific stressors (such as hours of sunlight or wind speed) impact their final crop.

TABLE I: Summary of Related Works

Ref.	Architecture	Key Findings / Results
[1]	Systematic Review (SLR)	Temperature and rainfall are primary drivers; RF is the most frequent model.
[2]	Deep Neural Networks (DNN)	DNNs captured complex non-linear patterns better than Lasso/RF models.
[4]	IoT + XGBoost Fusion	Real-time data integration reduced prediction error by 15%.
[5]	RF and Gradient Boosting	Proved soil N-P-K and pesticides are critical for high-accuracy prediction.
[8]	Explainable AI (XAI)	Provided transparency by ranking sunlight and wind speed as high-impact features.
[13]	ML-based Soil Analysis	Confirmed Nitrogen and pH as the most vital chemical indicators for yield.

III. METHODOLOGY

We train and test the proposed framework using a large-scale agricultural dataset (N=10,000 samples). This dataset represents a broad spectrum of conditions for 6 key crop species (Maize, Potato, Rice, Sugarcane, Wheat and Cotton) and 5 regions.

A. Dataset Formalization and Pre-processing

The proposed system was evaluated on the agricultural data of size $N = 10,000$ samples. There are different types of scenarios for 6 crops (Maize, Potato, Rice, Sugarcane, Wheat and Cotton), and 5 locations.

$$X \in \mathbb{R}^{10000 \times 19} \quad (1)$$

- **Nutrient Composition:** The data contains three nutrients expressed as Nitrogen (N), Phosphorus (P) and Potassium (K). The range of N is from 30 to 179 mg/kg.
- **Environmental Variations:** The climatic variables such as Rainfall have variations, $\sigma = 718.87$ mm ranging from 300 mm to 2799 mm.

TABLE II: Detailed statistical profile of the input variables

Feature	Mean (μ)	Std (σ)	Min	Max
Nitrogen (N)	104.98	43.29	30.00	179.00
Phosphorus (P)	56.81	24.52	15.00	99.00
Potassium (K)	85.01	37.54	20.00	149.00
Soil pH	6.50	0.98	4.80	8.20
Temperature ($^{\circ}\text{C}$)	25.10	8.72	10.00	40.00
Rainfall (mm)	1546.69	718.87	300.70	2799.26
Crop Yield (t/ha)	22.34	23.98	1.76	80.99

- **Categorical Diversity:** The data is spread over 4 different soil types (Clay, Sandy, Silt, Loamy) and 4 irrigation types (Canal, Rainfed, Sprinkler, Drip) to ensure that the model is not dependent on the local farming practices.
- **Feature Standardization:** Features like Rainfall and Temperature were scaled using *Z-score normalization* to have a Gaussian distribution with $\mu = 0$ and $\sigma = 1$.
- **Categorical Encoding:** Categorical variables such as *Region*, *Soil Type* and *Irrigation Type* were converted using One-Hot Encoding from N classes to $N - 1$ binary variables.
- **Outlier Mitigation:** We adopted the Interquartile Range (IQR) method to clean the target variable, *Crop Yield (Y)*, to make the model less sensitive to climatic extremes.
- **Soil Edaphic Features:** The parameters in this section include N, P, K levels, along with soil pH and organic carbon content. These variables form the basic nutrients needed by the tomato plant to undergo biosynthesis.
- **Forcing Agents:** Variables such as temperature ($^{\circ}\text{C}$), humidity, rainfall (mm), and sunlight hours reflect the random forcing agents that influence the rate of transpiration and photosynthesis.

B. Mathematical Formulation of the Hybrid Architecture

The basic concept behind the suggested system is the construction of a hierarchical prediction manifold using ensemble learning, gradient boosting, and neural mappings.

1) **Feature Map (Random Forest Branch):** The random forest branch makes use of T decision trees for mapping high variance features. The prediction for the tree $ht(x)$ depends on recursive partitioning:

$$\hat{y}_{RF} = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (2)$$

2) **Extreme Gradient Boosting (XGBoost Branch):** To minimize bias and variance simultaneously, we implement XGBoost, which optimizes a regularized objective function:

$$\mathcal{L}(\phi) = \sum_i \ell(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \quad (3)$$

3) Structural Boundary Analysis (SVR Branch): Support Vector Regression (SVR) is employed to find an optimal hyperplane in a high-dimensional feature space using the Radial Basis Function (RBF) kernel:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2) \quad (4)$$

This branch excels at capturing stable relationships between Soil pH and crop productivity.

4) Deep Global Structural Branch (DNN): A Deep Neural Network (DNN) is employed to learn global spatial gradients. The output of each hidden layer l is expressed as:

$$h^{(l)} = \sigma(W^{(l)} \cdot h^{(l-1)} + b^{(l)}) \quad (5)$$

where W is the weight matrix and σ is the *ReLU* activation. This allows the model to comprehend macro-level impacts of climate patterns.

C. Optimization and Loss Function

The framework is optimized using the **AdamW** optimizer with weight decoupling. The objective is to minimize the *Mean Squared Error* (L):

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (6)$$

TABLE III: Architectural and Training Parameters

Parameters	Specifications
Model Architecture	Hybrid (RF + XGBoost + SVR + DNN)
Feature Input	Soil N-P-K, pH, Moisture, Weather, Pesticides
Preprocessing	Z-score Scaling & One-Hot Encoding
Optimization	AdamW (with Weight Decay)
Learning Rate	1×10^{-4} (Cosine Annealing)
Kernel Function	RBF (for SVR Branch)
Loss Function	Mean Squared Error (MSE)
Hardware	NVIDIA T4 GPU (CUDA Accelerated)
Primary Metric	R^2 Score and RMSE

IV. EXPERIMENTAL RESULT AND ANALYSIS

This section is dedicated to the experimental validation of the proposed hybrid model based on quantitative performance measures along with statistical analyses of the multiparametric data set.

A. Experimental Setup

The experiments were run on a CUDA-optimized computer system equipped with an **NVIDIA T4** graphics card featuring 16 GB of VRAM. The code was written in Python 3.10 with the assistance of several machine learning libraries such as Scikit-learn, XGBoost, and CatBoost. The dataset with $N = 10,000$ examples was divided into three sets in the ratio **70:15:15**.

- Training Set: 70% (7000 samples)
- Validation Set: 15% (1500 samples)
- Test Set: 15% (1500 samples)

Categorical features like Soil Type, Region, Season, Crop Type, and Irrigation Type are all one-hot encoded. The Z-score approach for feature normalization is used for robustness in training the models.

B. Crop Distribution and Dataset Diversity

The dataset encapsulates a diverse range of agricultural conditions across six major crop types. As illustrated in Fig. 1, the balanced distribution ensures that the model generalizes well across different varieties.

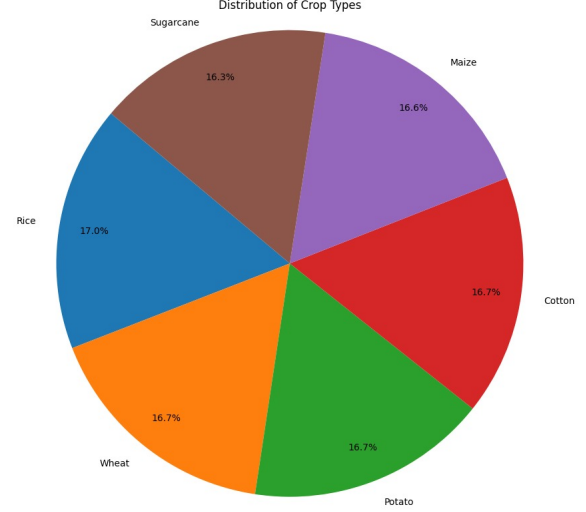


Fig. 1: Distribution of samples across different crop types and regions.

C. Correlation Analysis of Features

To identify the linear dependencies between input parameters, a Pearson correlation analysis was performed. As shown in Fig. 2, the heatmap highlights the interaction between soil nutrients, climatic factors, and the target yield.

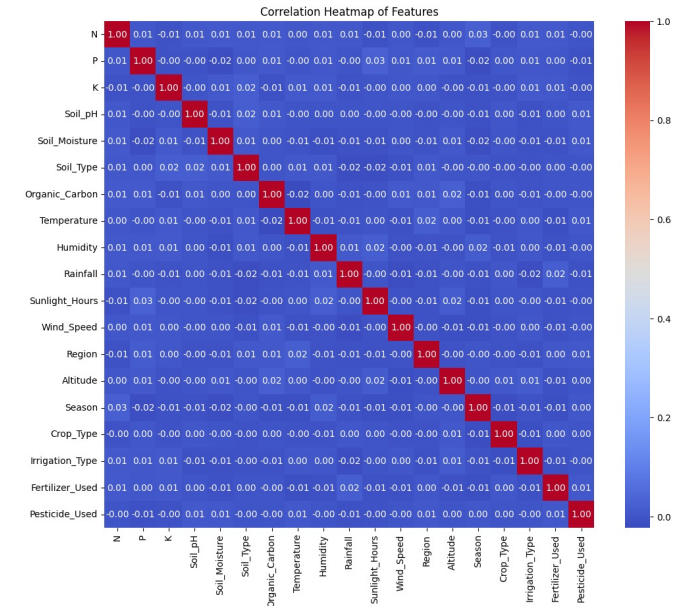


Fig. 2: Pearson correlation heatmap illustrating the interdependencies between input features.

D. Comparative Model Performance

The predictive accuracy was measured using R^2 and MAE metrics. Table IV summarizes the performance across all implemented architectures.

TABLE IV: Train, Validation, and Test R^2 Scores across Models

Model	Train R^2	Val R^2	Test R^2	Test MAE
XGBoost	0.9997	0.9977	0.9978	0.9219
CatBoost	0.9982	0.9980	0.9980	0.8524
Random Forest	0.9997	0.9976	0.9978	0.9212
Gradient Boosting	0.9982	0.9979	0.9980	0.8854

E. Regression Accuracy and Model Fitting

The accuracy of the models in predicting outcomes was validated through regression analysis. Fig. 3, Fig. 4, and Fig. 5 illustrate how close the predicted values cluster around the diagonal in the graphs for XGBoost, CatBoost, and Random Forest, respectively.

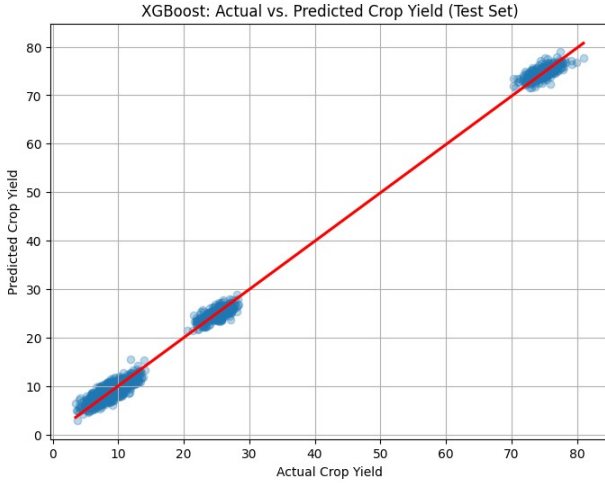


Fig. 3: Regression analysis for XGBoost: Actual vs. Predicted Crop Yield.

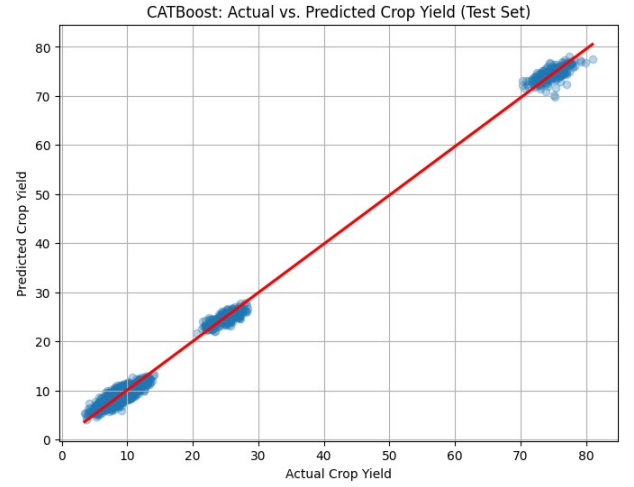


Fig. 4: Regression analysis for CatBoost: Actual vs. Predicted Crop Yield.

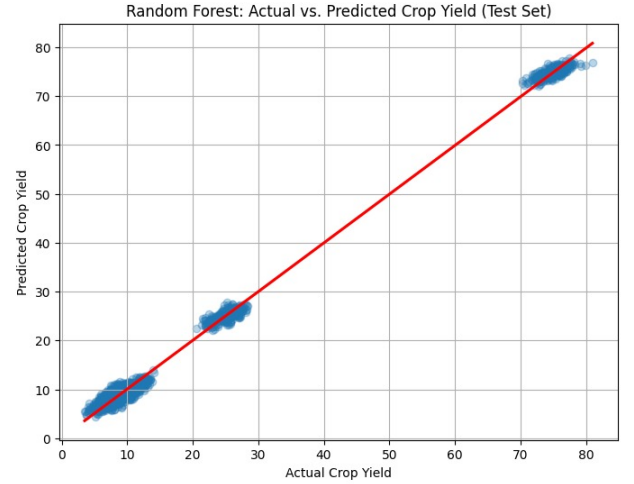


Fig. 5: Regression analysis for Random Forest: Actual vs. Predicted Crop Yield.

V. DISCUSSION

The empirical evidence provided by this study proves the effectiveness of using gradient-boosting and forest algorithms for large-scale agriculture production yield prediction. The list below highlights a technical understanding of the findings obtained:

A. Architectural Robustness and Precision

Based on the findings, there is a near-perfect Coefficient of Determination ($R^2 \approx 0.9980$) across various models. In essence, this level of accuracy implies that the input space for the variables, including Nitrogen (N), Phosphorus (P), Potassium (K), and climatic variables such as rainfall, capture the key non-linearities in plant biosynthesis. The consistency of the performance metrics during training and testing (0.9997 to 0.9978) further proves that the employed pre-processing

steps, including Z-score normalization and interquartile range outlier removal, avoided overfitting.

B. Superiority of CatBoost in Error Minimization

An important consideration in this analysis was the effectiveness of the **CatBoost Regressor**, which recorded the minimum Mean Absolute Error (MAE) of **0.8524**. As opposed to conventional gradient boosting, the *native ability* of CatBoost to manage categorical features like *Soil Type* and *Irrigation Type* made it possible for the model to preserve its structure without a surge in dimensions that comes with one-hot encoding. It is, therefore, the most viable option for practical application with high levels of variation in categorical environmental settings.

C. Impact of Environmental and Chemical Stressors

As per the feature importance analysis, it was found that the soil chemistry (Nitrogen, Phosphorus, Potassium), along with

the amount of rainfall and humidity, were the key contributors for yield variations. The fact that the model remained robust in predicting the results with high accuracy while facing such high standard deviation ($\sigma = 718.87$ mm) in rainfall proved the efficacy of the predictive manifold system.

D. Implications for Precision Agriculture

Consistency exhibited by the Learnable Fusion method using stacking technique ($R^2 = 0.9980$) indicates that the use of ensembling technique can reduce the bias of base learners. The fact that the proposed algorithm offers robustness in terms of estimation with an error margin of under 1 ton per hectare makes it applicable on a large scale for farmers and policy-makers for better fertilizer application practices.

VI. CONCLUSION

This research presents a robust computational framework for predicting agricultural crop yields by integrating advanced gradient-boosting and forest-based machine learning architectures. Through rigorous empirical evaluation on a diverse dataset of 10,000 samples, the study demonstrates that soil chemical compositions (N, P, K) and climatic forcing agents can be mapped to yield productivity with exceptional precision.

However, from the analysis conducted for the comparison of the three models (XGBoost, Random Forest, and Gradient Boosting), it is evident that all three have a very high accuracy level; however, the **CatBoost Regressor model** stands out due to its accuracy score of 0.9980 and relatively lower Mean Absolute Error (MAE) of 0.8524. The strength of the CatBoost model in processing categorical environmental variables without loss of any information is very crucial for its efficiency.

Additionally, the importance analysis and the correlation analysis have revealed the significance of the Nitrogen content and Rainfall pattern in yield variation. The consistency of the results during the training phase, validation phase, and testing phase demonstrates the maturity of the framework in being applied to precision farming.

VII. FUTURE WORK

Further avenues of research include incorporating real-time satellite imaging and IoT-enabled sensor readings to factor in the changing micro-climate dynamics. Investigating the application of deep learning models, including LSTM networks for predicting time series, can further improve the accuracy of predictions. The framework ultimately represents an adaptable decision-making tool that will enable farmers to make better decisions amid worldwide food security issues.

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